Cardiac Arrest

This guideline is for adults and children in non-traumatic cardiac arrest. See the **'cardiac arrest secondary to trauma'** guideline if the cardiac arrest is secondary to trauma.

Initial actions (the first 1-4 minutes)

- Perform continuous chest compressions while the defibrillator is being attached and charged.
- Defibrillate immediately if the cardiac rhythm is VF or VT using a single shock and immediately recommence chest compressions:
  - Maximum joule setting for an adult.
  - See the paediatric drug dose tables for joule settings for a child.
- Place an SGA and commence ventilation, preferably using a timing indicator.

Subsequent actions

- Perform two-minute cycles of high performance CPR between rhythm checks.
- Administer adrenaline IV every four minutes:
  - 1 mg IV for an adult.
  - See the paediatric drug dose tables for a child.
- Consider placing an ETT.
- Establish a team leader and utilise the **cardiac arrest checklist**.

If the rhythm is VF or VT

- Defibrillate using single shocks every two minutes.
- Immediately recommence chest compressions after each shock.
- If in VF or VT after three shocks, including shocks delivered prior to ambulance arrival:
  - Administer amiodarone:
    - 300 mg IV for an adult.
    - See the paediatric drug dose tables for a child.
  - Consider changing to anterior/posterior pad placement or performing double sequential defibrillation.
- If in VF or VT after six shocks, including shocks delivered prior to ambulance arrival, administer a second dose of amiodarone:
  - 150 mg IV for an adult.
  - See the paediatric drug dose tables for a child.

Cardiac Arrest

- Seek clinical advice if VF or VT persists.

If torsades de pointes is present or has occurred

- Withhold amiodarone.
- Administer magnesium IV:
  - 10 mmol (2.5 g) over 1-2 minutes for an adult.
  - See the paediatric drug dose tables for a child.
  - Repeat once if torsades de pointes recurs or persists.

If the patient is in PEA

- Correct immediately reversible causes.
- Administer 0.9% sodium chloride IV:
  - 1-2 litres IV for an adult.
  - 20-40 ml/kg IV for a child.

If patient movement is interfering with resuscitation

- Administer 1 mg/kg of ketamine IV, up to a maximum of 150 mg.
- This may be repeated once if required.
- If significant movement continues, consider administering 2 mg/kg of rocuronium IV up to a maximum of 200 mg, provided an ETT has been placed, and the position confirmed using $ETCO_2$.

Backup

- Backup from an ICP/CCP should be requested.

Cardiac arrest checklist

- ✓ Team leader identified.
- ✓ Roles allocated and confirmed.
- ✓ CPR feedback device and metronome on if available.
- ✓ Defibrillator screen visible to the team leader.
- ✓ Airway adjunct or ETT in place and ventilation adequate.

Cardiac Arrest

- ✓ $ETCO_2$ attached and waveform visible.
- ✓ Oxygen flow and supply adequate.
- ✓ IV/IO access secure, two sites if possible.
- ✓ Consider underlying and reversible causes.
- ✓ Appropriate medicines administered.
- ✓ Initiate documentation.
- ✓ Communication with family/bystanders.

Additional information

Deciding to start resuscitation

- Resuscitation should start unless there is a clear reason not to. Clear reasons for not starting resuscitation include:
  - » Signs of rigor mortis or post-mortem lividity.
  - » A clear advance directive not to receive resuscitation for cardiac arrest.
  - » Scenarios where resuscitation is futile or clearly not in the best interest of the patient. Examples include unwitnessed cardiac arrest with asystole as the initial rhythm, patients who are dying from cancer and patients with severe life-limiting disease.
- Ambulance personnel must consider all available information, including the patient's known wishes (for example, their advance directive) and act in what they believe is the best interest of the patient.
- If there is doubt regarding the appropriateness of a resuscitation attempt, resuscitation should begin while further information is gained.
- Nobody has the right to demand resuscitation on behalf of a patient, including those with an Enduring Power of Attorney and family/whānau, but the known wishes of the patient must be considered.
- Family/whānau do not have the right to decline resuscitation on behalf of a patient unless they have an Enduring Power of Attorney. See the **'treatment and/or transport without informed consent'** guideline for more information.
- There must be clear documentation regarding decisions made.

Cardiac Arrest

Deciding to stop resuscitation

- There is no absolute time at which it is possible to say that further resuscitation is futile. Stopping resuscitation requires clinical judgement on the likelihood of survival, considering the following:
  - The cause of the cardiac arrest.
  - Whether or not the cardiac arrest was witnessed.
  - Whether or not there was bystander CPR.
  - The response time.
  - The initial rhythm.
  - The total estimated time in cardiac arrest.
  - The end tidal CO_2.
  - Whether ROSC has occurred at any time.
  - The patient's comorbidities.

Deciding to stop resuscitation: EMTs

- Resuscitation must continue until a Paramedic, ICP or CCP arrives and makes a decision to stop, or advice is received via the Clinical Desk that it is appropriate to stop. If backup is not available, and the Clinical Desk cannot be contacted:
  - If the arrest was unwitnessed and no shock is advised, the prognosis is very poor and it is appropriate to stop resuscitation if there are no signs of ROSC 20 minutes after the onset of resuscitation by ambulance personnel.
  - For other circumstances it is appropriate to stop resuscitation if there are no signs of ROSC 40 minutes after the onset of resuscitation by ambulance personnel.

Deciding to stop resuscitation: Paramedics, ICPs and CCPs

- It is usually appropriate to stop resuscitation:
  - 20 minutes after the onset of resuscitation by ambulance personnel in poor prognosis scenarios.
  - 40 minutes after the onset of resuscitation by ambulance personnel in good prognosis scenarios.
  - Earlier than described above, if it becomes clear that it was inappropriate to have commenced resuscitation, or the rhythm has deteriorated into asystole for more than a few minutes despite resuscitation.

Cardiac Arrest

Movement during cardiac arrest

- The patient may move during cardiac arrest if cerebral blood flow is being maintained by very good CPR.
- This can cause personnel to believe that the patient is not in cardiac arrest and may lead to delayed defibrillation and/or inappropriate pauses in CPR.
- It is appropriate to briefly pause CPR to confirm the patient is in cardiac arrest.
- If rocuronium is administered, neuromuscular blockade will continue if sustained ROSC is achieved and the patient will then require an adequate level of sedation.

The prognosis of cardiac arrest

There is no one factor that can be used to determine the prognosis of an individual patient in cardiac arrest. Multiple factors must be considered, noting that in most patients there will be a mixture of prognostic factors.

Cardiac Arrest

Poor prognostic factors

- Secondary cardiac arrest
- Unwitnessed
- No bystander CPR
- Response time > 8 minutes
- Initial rhythm asystole or PEA
- Time in cardiac arrest > 30 minutes
- Severe comorbidities
- Living in aged residential care
- Age > 80 years
- $ETCO_2$ < 15 mmHg or falling despite CPR

Good prognostic factors

- Primary cardiac arrest
- Witnessed
- Bystander CPR
- Response time < 8 minutes
- Initial rhythm VT or VF
- Time in cardiac arrest < 30 minutes
- No severe comorbidities
- Living independently
- Age ≤ 80 years
- $ETCO_2$ > 25 mmHg with CPR

- Severe comorbidities are chronic diseases that significantly limit a patient's life expectancy. Examples include dementia, severe COPD, severe heart failure, a requirement for aged residential care and metastatic cancer with weight loss.

- The reactivity of the patient's pupils during and immediately following cardiac arrest is not prognostic and there is no role for including pupil reactivity in decision making.

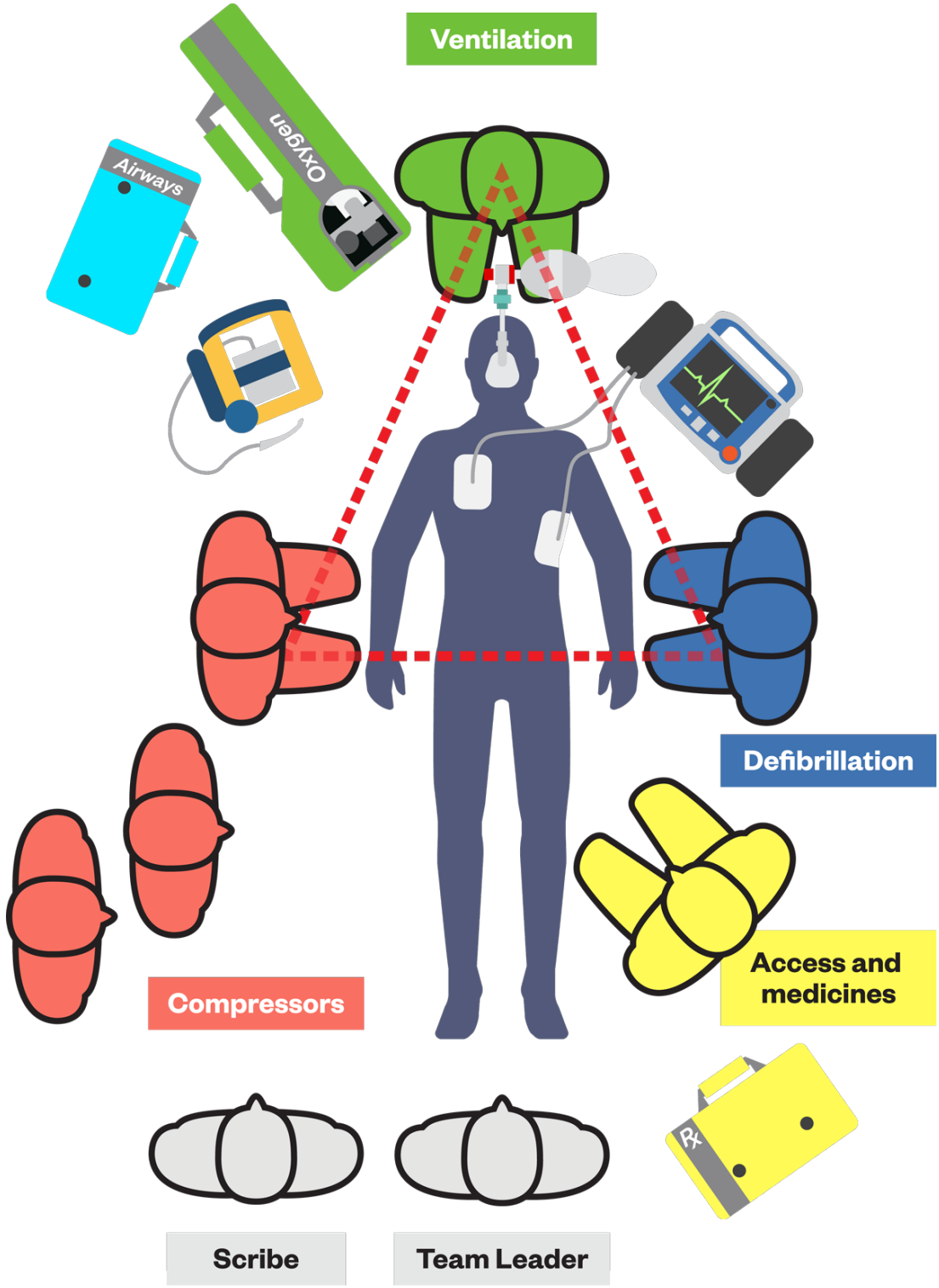
High performance CPR

- High performance CPR encompasses the concepts of:
  - Optimising chest compression depth, rate, and recoil.

- Minimising interruptions to chest compressions.
- Avoiding over-ventilation.

- Perform chest compressions at a rate of 100-120/minute, ensuring adequate depth and complete recoil.
- CPR should be performed for two minute cycles between rhythm checks. The person performing chest compressions should ideally change every two minutes, or earlier if tired.
- A CPR feedback device and/or metronome must be used if available.
- Pauses in CPR for rhythm checks, defibrillation or ventilation should be minimised, and usually last no longer than three seconds. A pause of up to 10 seconds is permitted for a pulse check in PEA.
- The person performing chest compressions should hover approximately 5 cm above the patient's chest during the pause.

Cardiac Arrest



Ventilation in high performance CPR

- Ventilation has a low priority during the first few minutes of resuscitation if the patient has had a primary cardiac arrest.

Cardiac Arrest

- Once ventilation has commenced, the ventilation rate should be no more than 10/minute, as rates higher than this increase intrathoracic pressure and therefore reduce venous return during CPR.
- When ventilation is provided via a bag and mask:
  - For an adult the CPR compression to ventilation ratio is 30:2. This ratio prioritises chest compressions on the basis that an adult is most likely to have had a primary cardiac arrest. If an adult has had a cardiac arrest secondary to asphyxiation or respiratory failure, alter the ratio to 15:2.
  - For a child the CPR compression to ventilation ratio is 15:2 (exception – the ratio is 3:1 for newborns). The 15:2 ratio reduces the priority of chest compressions on the basis that a child is most likely to have had a cardiac arrest secondary to respiratory failure. If a child has had a primary cardiac arrest, alter the ratio to 30:2.
- When ventilation is being provided via an SGA or ETT:
  - Perform continuous chest compressions and ventilate adults and children at a rate of 10/minute, preferably using a ventilation timing light.
  - If continuous chest compressions clearly impair ventilation via an SGA, interrupt the chest compressions to provide ventilations using the ratios described above.
- Intubation is not a high priority during the initial management of cardiac arrest. If ROSC has not been achieved within 10 minutes, intubation with an ETT may occur provided this does not interrupt chest compressions.

IV access and medicines

- Gaining IV access and administering medicines has a lower priority than providing high performance CPR.
- IV access is preferred over IO access. Clinical judgement should be used and IO access should usually be obtained if IV access cannot be gained, and resuscitation is ongoing.
- All doses should be flushed with a running line of 0.9% sodium chloride or a minimum of 20 ml 0.9% sodium chloride.

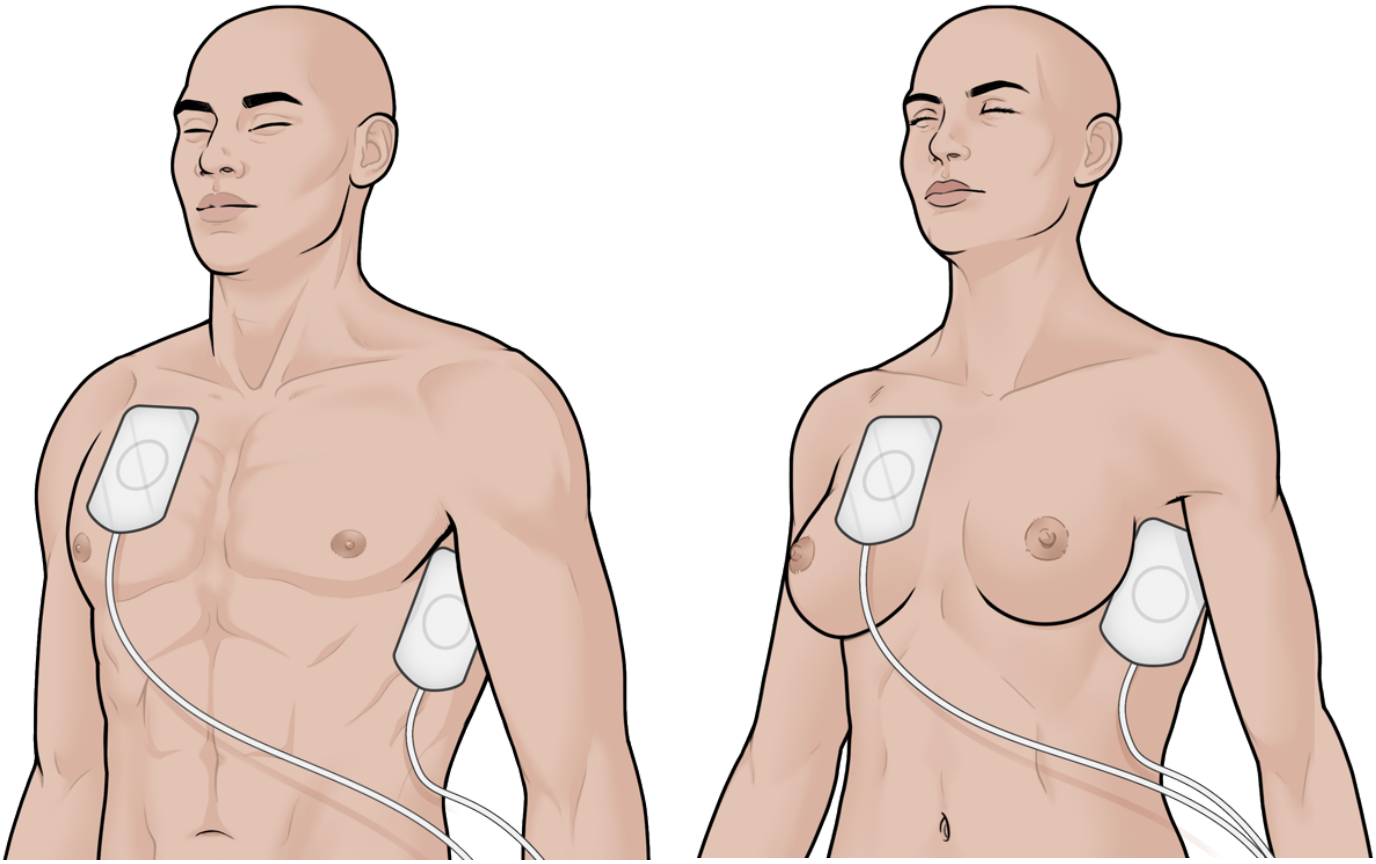
End tidal CO_2

- End tidal CO_2 ($ETCO_2$) is a marker of the blood flow being achieved during CPR. With good CPR an $ETCO_2$ of greater than 20 mmHg should usually be achieved, unless there is an obstruction to blood flow, for example from a pulmonary embolus or tension pneumothorax.
- $ETCO_2$ must be used to confirm ETT placement and must be continuously measured if the patient has been intubated with an ETT.

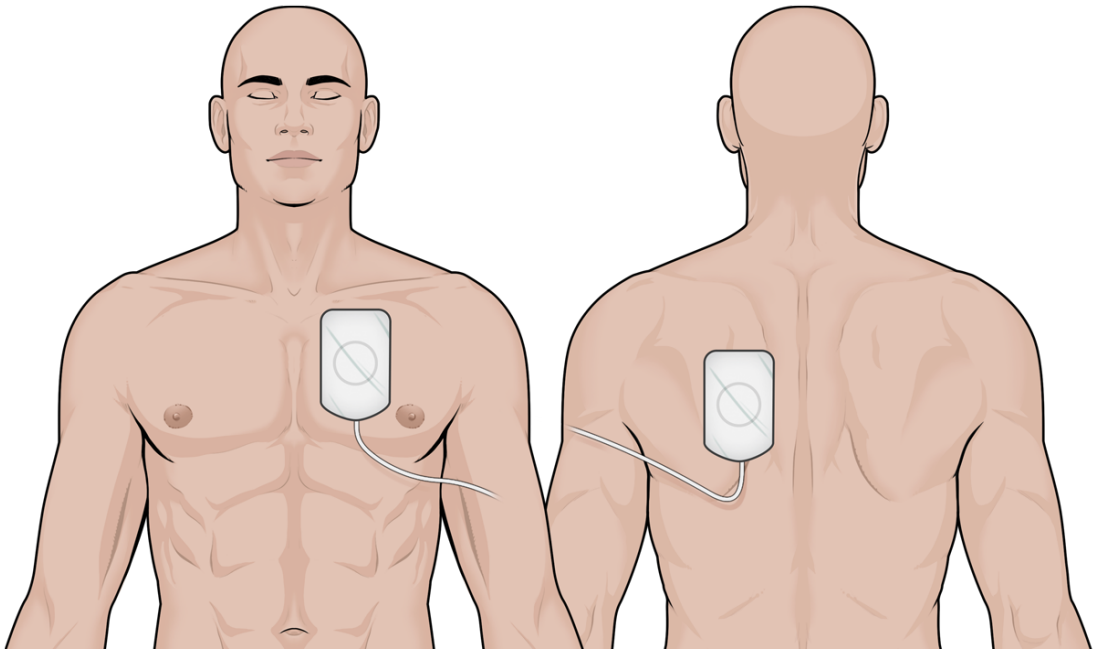
- $ETCO_2$ should be continuously measured if the patient is being ventilated via an SGA, noting that the $ETCO_2$ will be reduced if there is significant leak around the cuff. In this setting the trend of the $ETCO_2$ is more important than individual recordings.
- $ETCO_2$ may be measured if the patient is being ventilated via a face mask, noting that the $ETCO_2$ will be reduced if there is significant leak around the mask. In this setting the trend of the $ETCO_2$ is more important than individual recordings.
- A sudden increase in $ETCO_2$ is usually an indicator that ROSC has occurred.
- A falling $ETCO_2$ despite CPR is a poor prognostic factor and must prompt a focus on ensuring high performance CPR is being provided and the ventilation rate is at or below 10/minute.
- An $ETCO_2$ of less than 15 mmHg despite CPR is a very poor prognostic factor.

Defibrillation

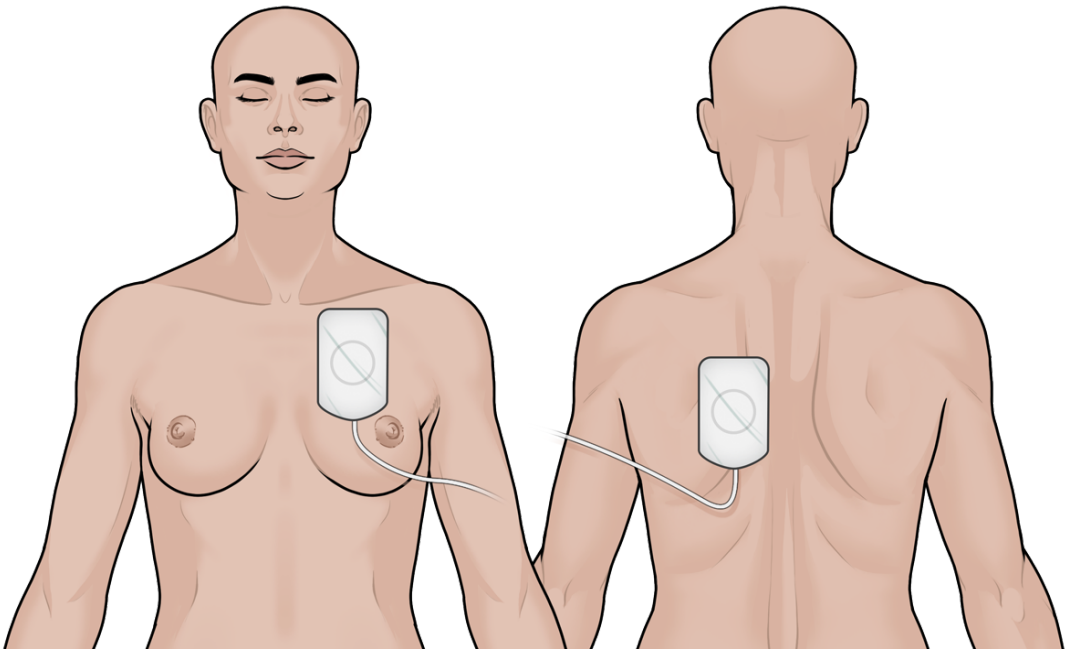
- Pad placement and adherence is important. It is worth taking the extra time to ensure the pads are placed in the ideal location and are well adhered.
- Routinely place pads in the anterior/posterior position for small children and morbidly obese patients.
- If VF or VT persists or recurs after three shocks, consider whether it is recurrent, or refractory. Because CPR is initiated immediately after each shock, it is not always possible to differentiate:
  - VF or VT that is successfully reverted with each shock and then recurs, and thus is recurrent, versus
  - VF or VT that is not successfully reverted with each shock, and thus is refractory.
- If VF or VT is clearly recurrent, changing the defibrillation strategy is not required.
- If VF or VT is clearly or possibly refractory, the defibrillation strategy should be changed.
- The two options when changing the defibrillation strategy are:
  - Anterior/posterior pad placement, or
  - Double sequential defibrillation (DSD).
- Anterior/posterior pad placement is the preferred approach when one manual defibrillator is present, or if the team are not fully trained in DSD.
- DSD is the preferred approach when two manual defibrillators are present, and the team is fully trained in DSD.



Antero-lateral pad placement for defibrillation



Cardiac Arrest

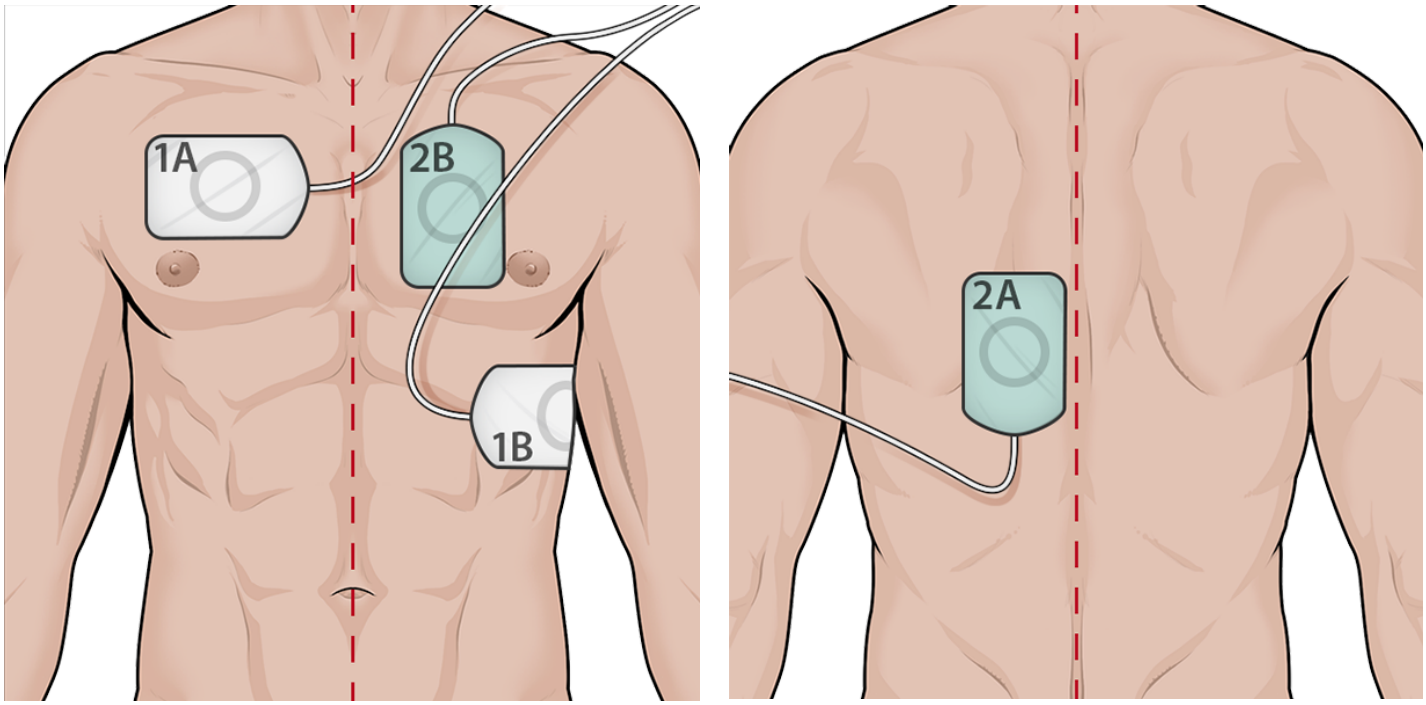


Antero-posterior pad placement for defibrillation

Double sequential defibrillation

- Double sequential defibrillation (DSD) is the provision of two sequential (approximately one second apart) shocks, using two manual defibrillators, with the pads attached in two different vectors, one in the antero-lateral vector and the other in the antero-posterior vector.
- When providing DSD:
  - Only those personnel specifically trained in the technique may provide DSD.
  - Use only manual defibrillators.
  - Ensure none of the pads are touching.
  - One person must discharge both defibrillators sequentially, to avoid simultaneous defibrillation.

Cardiac Arrest



Pad placement for double sequential defibrillation

Pulseless electrical activity (PEA)

- PEA is present when a patient in cardiac arrest has a rhythm that should be associated with cardiac output but is not. PEA is a clinical condition and not an abnormal rhythm.
- PEA is often secondary to a non-cardiac problem. The history immediately prior to cardiac arrest is very important in helping to determine what the cause may be.
- Potentially reversible causes of PEA include:
  - Hypoxia.
  - Hypothermia.
  - Hypovolaemia (including anaphylaxis).
  - Hyper/hypokalaemia (and other metabolic abnormalities).
  - Tension pneumothorax.
  - Tamponade (cardiac).
  - Toxins (poisoning).
  - Thrombosis (pulmonary and coronary).

Cardiac Arrest

- Some references include hydrogen ions (acidosis) and hypoglycaemia in the list of causes of PEA. However, neither of these cause cardiac arrest in the absence of another clinical problem, even when severe. Hypoglycaemia should be ruled out following ROSC.
- Consideration should be given to transporting the patient to hospital with CPR en route if a potentially reversible cause is identified that cannot be treated at the scene. To be successful, such a decision must be made very early in the resuscitation attempt.
- It is common for PEA to degenerate into asystole with time. During this process slow (less than 30/minute) broad complexes may be present. This is treated as asystole and not PEA.
- Survival is rare if the rhythm deteriorates from PEA into asystole despite resuscitation and prolonged resuscitation attempts in this setting are usually inappropriate.
- In general, patients that survive cardiac arrest with an initial rhythm of PEA do so because the underlying problem is immediately identified and corrected.
- If the patient remains in PEA despite resuscitation attempts, consider the possibility that there is cardiac output that cannot be detected clinically. In this setting stop chest compressions and observe the $ETCO_2$ for one minute while ventilating at a rate of 10/minute:
  - If the $ETCO_2$ rapidly falls to below 10 mmHg the patient is in cardiac arrest.
  - If the $ETCO_2$ is maintained at or above 10 mmHg, the patient has low cardiac output and should be treated accordingly.
- If available, ultrasound can be useful to assess for cardiac activity and reversible causes of PEA arrest.

Asystole

- Confirm the rhythm is asystole by checking:
  - The cables.
  - The leads.
  - Which lead is selected on the screen.
  - The amplitude.
- Exclude bradycardia which can look like asystole at a glance, particularly in children.
- Survival from cardiac arrest with an initial rhythm of asystole is rare and prolonged resuscitation attempts in this setting are usually inappropriate, particularly if the cardiac arrest was unwitnessed.

Single responder resuscitation

Cardiac Arrest

- Attach and use a defibrillator immediately. It is acceptable to use the defibrillator in automatic mode until additional personnel arrive.
- Perform CPR with a focus on chest compressions, utilising bystanders to help.
- Interventions such as IV access or intubation are not a priority and should only occur after additional trained personnel arrive.

Transport to hospital with CPR en route

- Transport to hospital with CPR en route should be rare because:
  - There is no convincing evidence it improves patient outcomes.
  - The quality of the resuscitation attempt (particularly the quality of CPR) is usually compromised, unless mechanical CPR is available.
  - Unrestrained personnel are at risk in the back of a moving ambulance.
- However, a small number of patients may benefit from transport to hospital with CPR en route, particularly if there is a potentially reversible cause that ambulance personnel are unable to treat (for example, hypothermia and poisoning). To be successful, such a decision must be made very early in the resuscitation attempt.
- Provide as much pre-hospital warning as possible.

Response and/or transport by helicopter

- It is uncommon for a helicopter to locate a patient within a timeframe that provides meaningful clinical benefit.
- Response by helicopter may be indicated if the patient is suspected to be in cardiac arrest and the helicopter will arrive on scene:
  - Prior to a road ambulance resource and a resuscitation attempt is expected, or
  - To deliver a CCP skill that will likely improve the patient's outcome (for example, cardiac arrest secondary to drowning).

Mechanical CPR

- Mechanical CPR does not usually have a role early in cardiac arrest but should be considered when:
  - It is not safe to perform manual chest compressions, for example in a moving ambulance.
  - There is insufficient room or personnel to perform chest compressions.
  - A prolonged resuscitation attempt is anticipated.

Cardiac Arrest

- One trained person must be present to supervise the mechanical CPR device during use.

Defibrillator failure action list

- Use this action list if a defibrillator fails and there is not another defibrillator (including an AED) immediately available. At each defibrillator intervention, pause briefly to determine if the problem has been fixed.
  - Task specific personnel to focus on resuscitating the patient.
  - Task specific personnel to focus on troubleshooting the defibrillator.
  - Ensure the pads are attached and connected.
  - Ensure all components of the defibrillator are physically connected.
  - Ensure the ECG leads are attached.
  - Change the lead shown on the screen so that the rhythm is visible.
  - Attach and connect a new set of pads.
  - Switch to automatic mode if in manual mode.
  - Turn the defibrillator off for 30 seconds and turn it back on again.
  - Seek urgent clinical advice if the situation is not resolved.

Cardiac arrest in special situations

Cardiac arrest secondary to drowning

- Prioritise the ventilation aspect of CPR and use a CPR ratio of 15:2 unless an ETT is in place.
- Place an ETT if ROSC is not achieved in the first few minutes.
- IV medicines have a very low priority.
- See the **'drowning'** guideline for additional information.

Cardiac arrest secondary to hanging

- Prioritise the ventilation aspect of CPR and use a CPR ratio of 15:2 unless an ETT is in place.
- Cervical spine immobilisation is not routinely required. This is because clinically significant cervical spine injury following hanging is extremely rare unless the patient has fallen the height of their body.
- The survival rate is low, but survivors usually come from the group that are in PEA and get ROSC within 5-10 minutes with good CPR alone.

Cardiac Arrest

- Prolonged resuscitation in the presence of asystole is inappropriate.
- IV medicines have a very low priority.

Cardiac arrest secondary to asthma

- Focus on using a ventilation rate of only 6/minute to avoid dynamic hyperinflation (gas trapping).
- IV adrenaline has a high priority.
- Exclude tension pneumothorax, noting this is rare. Needle chest decompression carries a significant risk of causing pneumothorax and finger thoracostomy is the preferred technique if chest decompression is required.

Cardiac arrest secondary to anaphylaxis

- IV adrenaline has a high priority.
- If the patient is in PEA and not immediately responding to resuscitation, escalate the adrenaline doses:
  - For an adult escalate the second dose to 3 mg, the third dose to 5 mg and then revert to standard 1 mg dosing.
  - For a child escalate the doses following the same principle.
- 0.9% sodium chloride IV has a high priority:
  - For an adult administer 2-3 litres of 0.9% sodium chloride IV.
  - For a child administer 40-60 ml/kg of 0.9% sodium chloride IV.

Cardiac arrest secondary to cyclic antidepressant poisoning

- The cardiac toxicity of cyclic antidepressants is partly caused by blockade of sodium channels within the heart and may be reduced by a large bolus of sodium ions.
- Administer 8.4% sodium bicarbonate IV:
  - 100 ml for an adult.
  - 2 ml/kg for a child.
- Administer 0.9% sodium chloride IV:
  - 1-2 litres for an adult.
  - 20-40 ml/kg for a child.

Cardiac Arrest

- Do not administer amiodarone because it can be associated with severe worsening of shock without resolution of the rhythm.

Cardiac arrest during pregnancy greater than 20 weeks

- In pregnancy where gestation is greater than 20 weeks, the uterus may impede venous return through the inferior vena cava in the supine position. Manually displace the uterus to the left to alleviate this.
- Consider transporting the patient with CPR en route (focusing on good chest compressions) if ROSC is not immediately achieved and time to a hospital is less than 10 minutes. Provide as much pre-hospital warning as possible. In this setting the primary reason for emergency caesarean section is to improve the chance of survival for the mother.

Cardiac arrest secondary to hypothermia

- See the **'hypothermia'** guideline.

Cardiac arrest secondary to hyperkalaemia

- See the **'hyperkalaemia'** guideline.

Cardiac arrest secondary to opiate poisoning

- The best treatment is CPR that includes a focus on ventilation.
- There is no role for naloxone because cardiac arrest is secondary to respiratory arrest and once cardiac arrest has occurred naloxone has no useful effect. If ROSC occurs, naloxone should still not be administered because it may be associated with seizures, hypertension, pulmonary oedema or severe agitation.

Cardiac arrest and implanted defibrillators/pacemakers

- Implanted defibrillators and pacemakers are usually situated in the soft tissue under the left clavicle.
- Place defibrillation pads at least 8 cm from the implanted device if possible and consider utilising the anterior/posterior position.

Cardiac Arrest

Cardiac arrest occurring in infants during sleep

- Prioritise the ventilation aspect of CPR using a CPR ratio of 15:2.
- Beware of misdiagnosing severe bradycardia as asystole.
- Survival is very low, but survivors tend to come from the group of patients that get ROSC within 5-10 minutes with good CPR alone.
- Prolonged resuscitation in the presence of asystole is inappropriate.
- IV medicines have a very low priority.
- Transport to hospital with CPR en route is inappropriate.

More Information

Version history

16 June 2026 1.1.0.1
Update: Resuscitation guidance updated.

11 September 2023 1.0.4
Update: Multiple changes.